Machine Learning-Based Model Accurately Identifies the Probability of Choledocholithiasis and Provides a Personalized Management Plan

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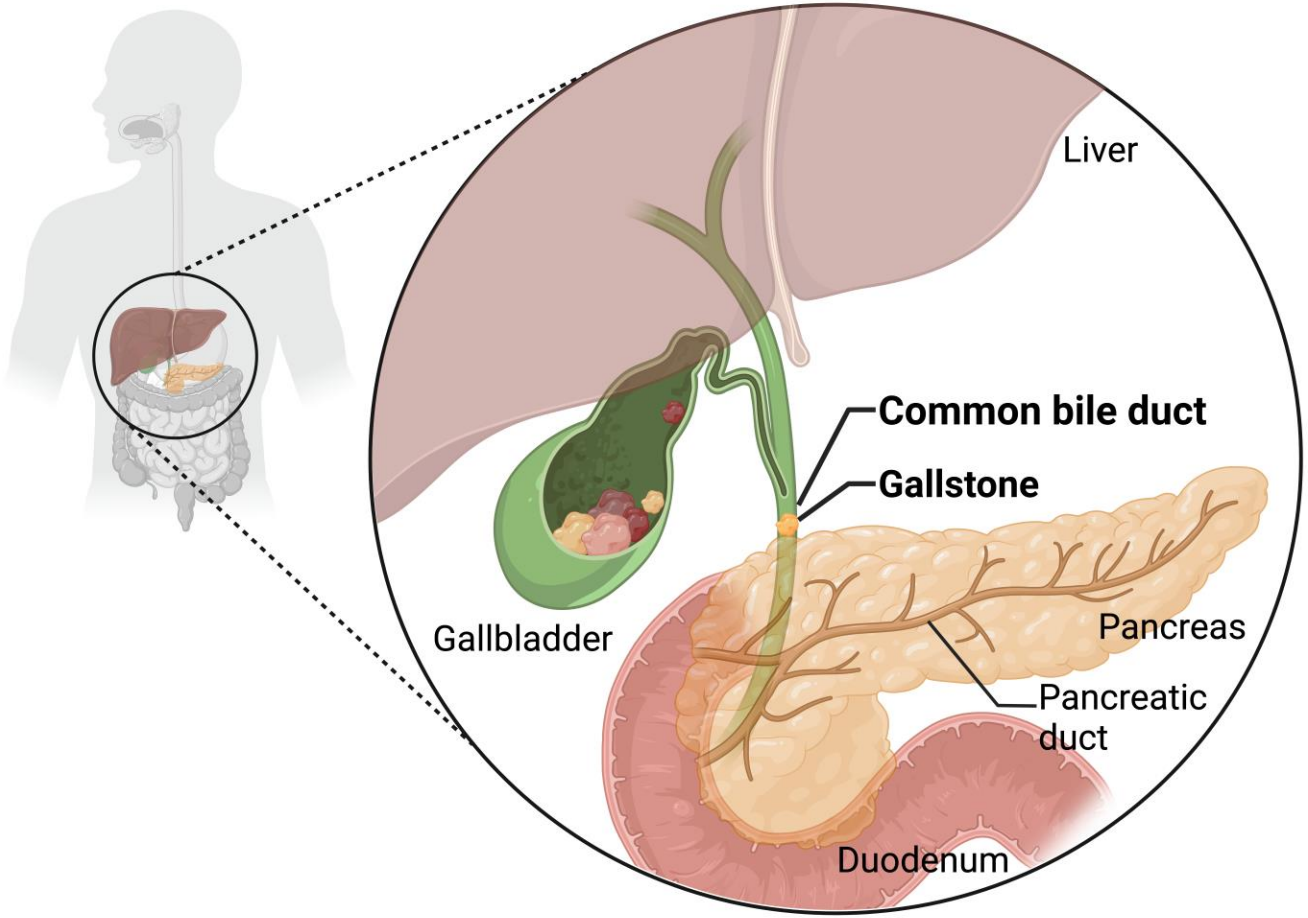
Outline

- Accurate identification of choledocholithiasis prevents morbidity and mortality.
- Current identification methods are suboptimal.
- Machine learning models can accurately predict choledocholithiasis.
- These models can guide patients to appropriate interventions and improve outcomes.

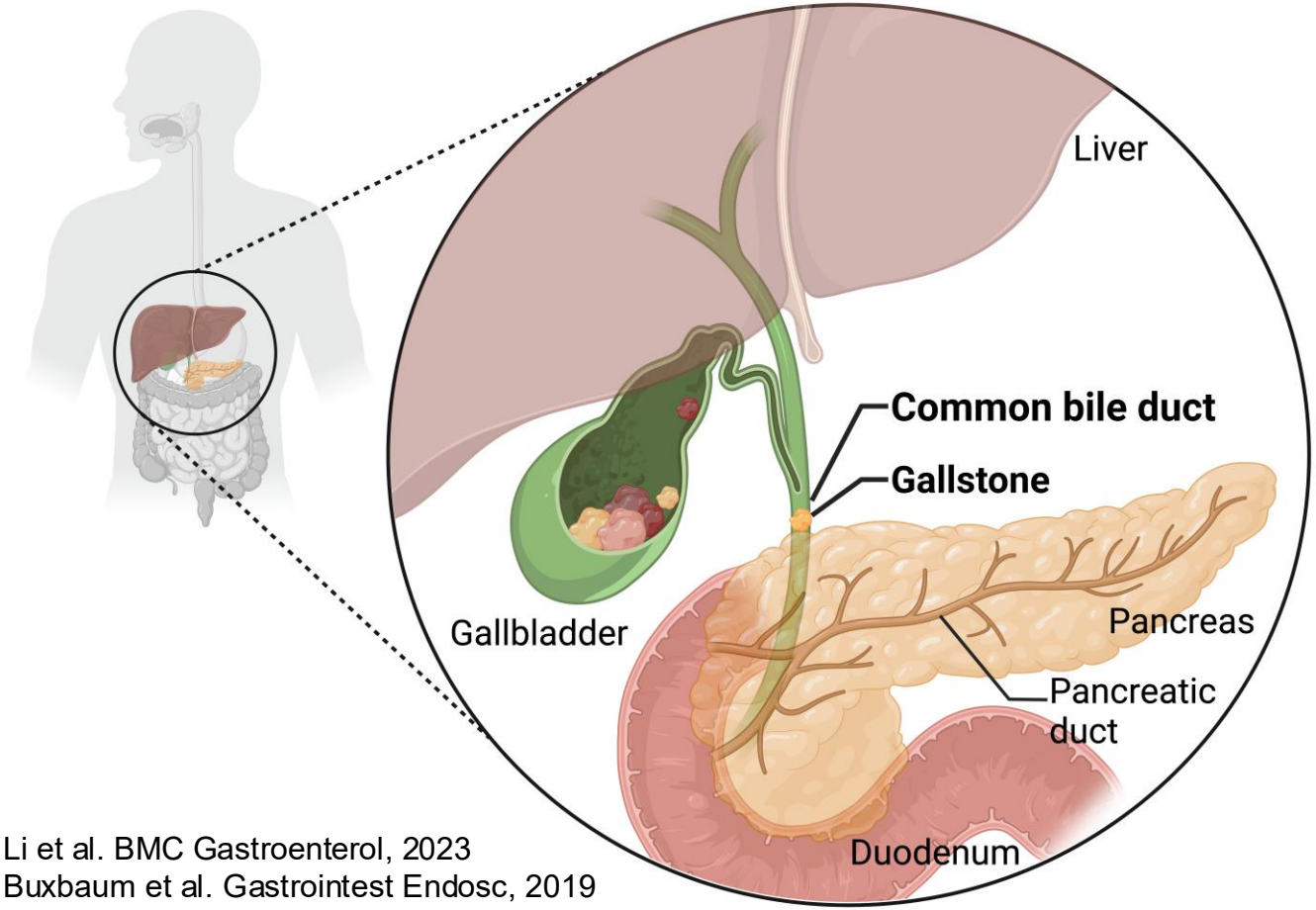
Choledocholithiasis causes morbidity and mortality



- 193 million patients experience biliary disease globally[1]
  - 4 million disability life years
  - $6.2 billion/year in the US
- Choledocholithiasis causes the majority of morbidity and mortality[2]
  - Biliary obstruction
  - Cholangitis
  - Biliary pancreatitis
  - Death

1. Li et al. World J Gastrointest Surg, 2023
2. Li et al. BMC Gastroenterol, 2023

Morbidity and mortality can be prevented



- CBD stone removal decreases mortality by 20%[3]
  - Accomplished with ERCP
- Unnecessary interventions carry 5-15% complication rate[2] and >$4151 cost[4]
- Cost effectiveness analyses with avoidance of unnecessary procedures:
  - $3919 - $8629 cost savings/procedure[5]
  - 1-3 days decreased length of stay[5]
  - $156,805 cost savings/hospitalization[6]
- Accurate identification is crucial for appropriate intervention

2. Li et al. BMC Gastroenterol, 2023
3. Buxbaum et al. Gastrointest Endosc, 2019
4. Procedure Price Lookup for Outpatient Services | Medicare.gov
5. Ali et al. Gastrointestinal Endoscopy, 2022
6. Kelley et al. The American Surgeon, 2025

Current identification methods are suboptimal

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Low risk of choledocholithiasis

- No risk factors

Intermediate risk of choledocholithiasis

- Abnormal liver enzymes
- Age > 55 years
- Bile duct dilation

High risk of choledocholithiasis

- CBD stone on imaging
- Tbili > 4 & bile duct dilation
- Ascending cholangitis

Cholecystectomy ± IOC ← MRI/MRCP (or EUS) ← (EUS &) ERCP

70-75% accuracy in validation studies[3,7,8]

13%-24% unnecessary ERCPs[7,8,9,10]

CBD = common bile duct, ERCP = endoscopic retrograde cholangiopancreatography, EUS = endoscopic ultrasound, IOC = intraoperative cholangiogram, MRCP = magnetic resonance cholangiopancreatography, Tbili = total bilirubin

3. Buxbaum et al. Gastrointest Endosc, 2019
7. Hasak et al. Surg Endosc, 2022
8. Wangchuk & Pongsakorn, Surg Endosc, 2022
9. Silva-Santisteban et al. Endosc Int Open, 2023
10. Chandran et al. Gastrointest Endosc, 2021

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Aim: develop a decision-making tool using machine learning model algorithms

1. Accurately predict the risk of choledocholithiasis
2. Guide appropriate intervention selection

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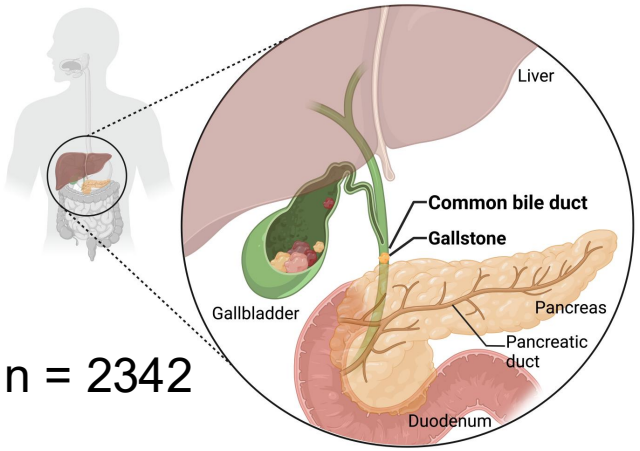
Decision-making tool creation

Suspected choledocholithiasis



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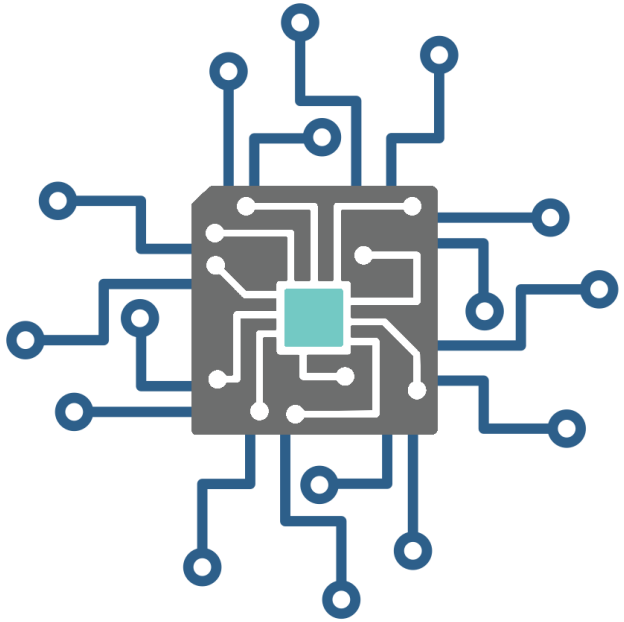
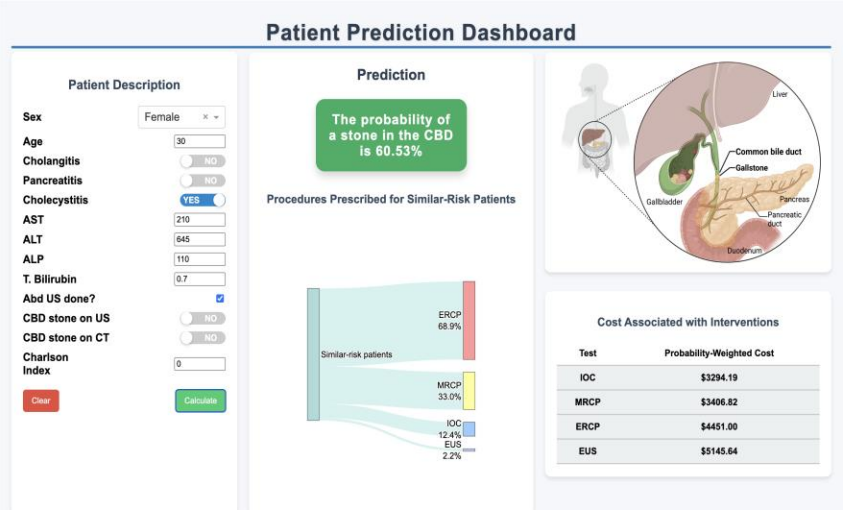




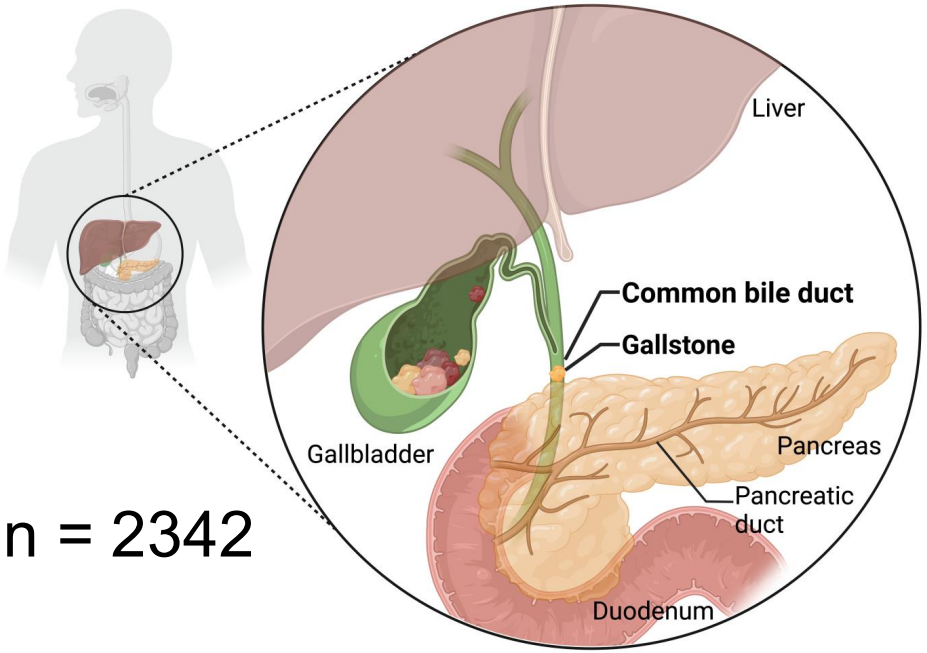
Database creation

Prediction Model Creation

User Interface Creation

Database Creation

Suspected choledocholithiasis: right upper quadrant abdominal pain, elevated liver enzymes, and imaging evidence of biliary ductal dilation or biliary stones, and/or ascending cholangitis



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Baseline Characteristics					
Factor	Choledocholithiasis		No Choledocholithiasis		P-value
N (n, % of total)	1103	46%	1282	54%	
Age (Avg, SD)	58.7	20	54.8	17.4	<0.0001
Female sex (n, %)	663	60%	800	62%	0.2514
Cholangitis (n, %)	292	26%	43	3%	<0.00001
Pancreatitis (n, %)	154	14%	44	3%	<0.00001
Cholecystitis (n, %)	263	24%	122	10%	<0.00001

Choledocholithiasis confirmed by visualization on advanced imaging, endoscopic procedure, or intraoperative cholangiogram or clinical follow-up

Avg = average, SD = standard deviation

Database Creation

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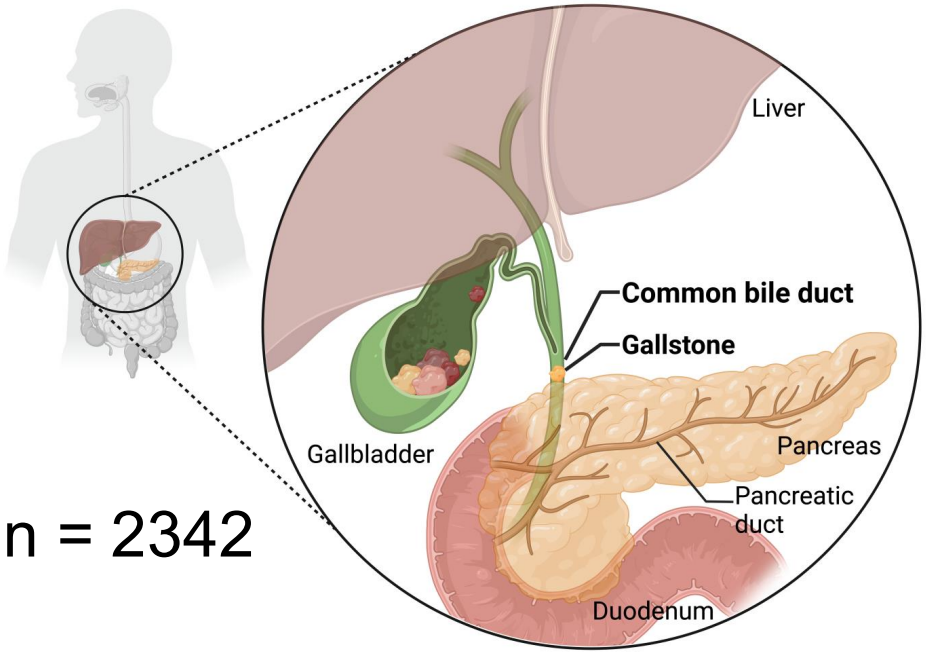

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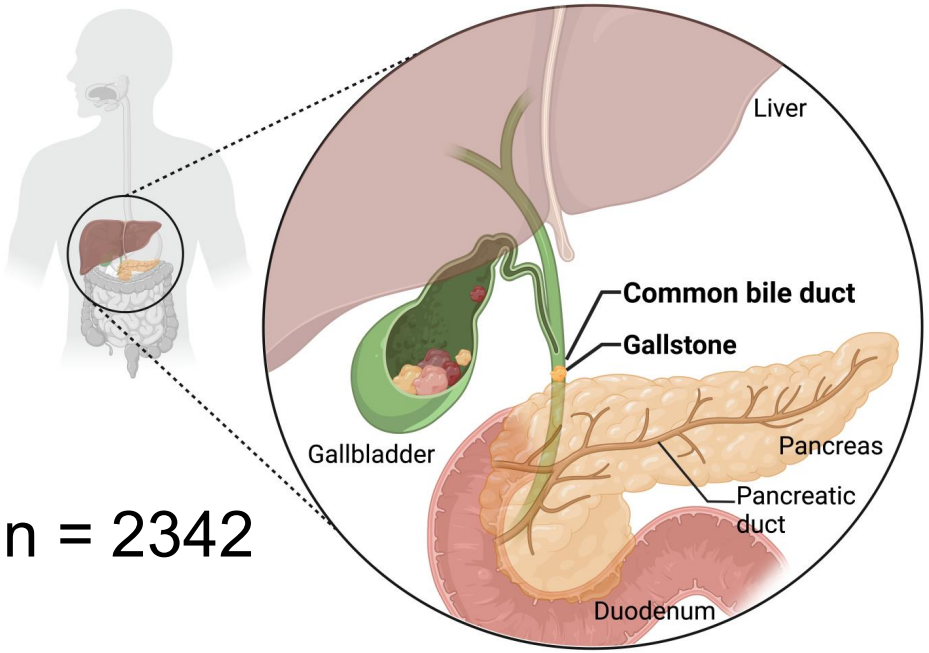
Laboratory					
	Choledocholithiasis		No Choledocholithiasis		P value
Day 0					
AST (Avg, SD)	232.6	351.1	82.9	274.0	<0.0001
ALT (Avg, SD)	241.6	301.3	85.2	188.6	<0.0001
Alk Phos (Avg, SD)	256.4	241.1	153.3	255.3	<0.0001
Total Bilirubin (Avg, SD)	3	3.8	1.4	3.3	<0.0001
Day 1					
AST (Avg, SD)	234.2	117.7	232.7	328.7	0.8856
ALT (Avg, SD)	278.4	46.9	182.2	135.4	<0.0001
Alk Phos (Avg, SD)	274.1	59	224.1	39.4	<0.0001
Total Bilirubin (Avg, SD)	3.8	0.5	3.1	0.6	<0.0001
Day 2					
AST (Avg, SD)	132.2	133.4	139.6	343.2	0.5006
ALT (Avg, SD)	221.4	224	171.3	316.6	<0.0001
Alk Phos (Avg, SD)	246.5	217.5	224.5	244.4	0.0212
Total Bilirubin (Avg, SD)	3.1	3.8	2.8	5.1	0.1082

Alk Phos = alkaline phosphatase, ALT = alanine aminotransferase, AST = aspartate aminotransferase, Avg = average, SD = standard deviation

Database Creation

Suspected choledocholithiasis: right upper quadrant abdominal pain, elevated liver enzymes, and imaging evidence of biliary ductal dilation or biliary stones, and/or ascending cholangitis



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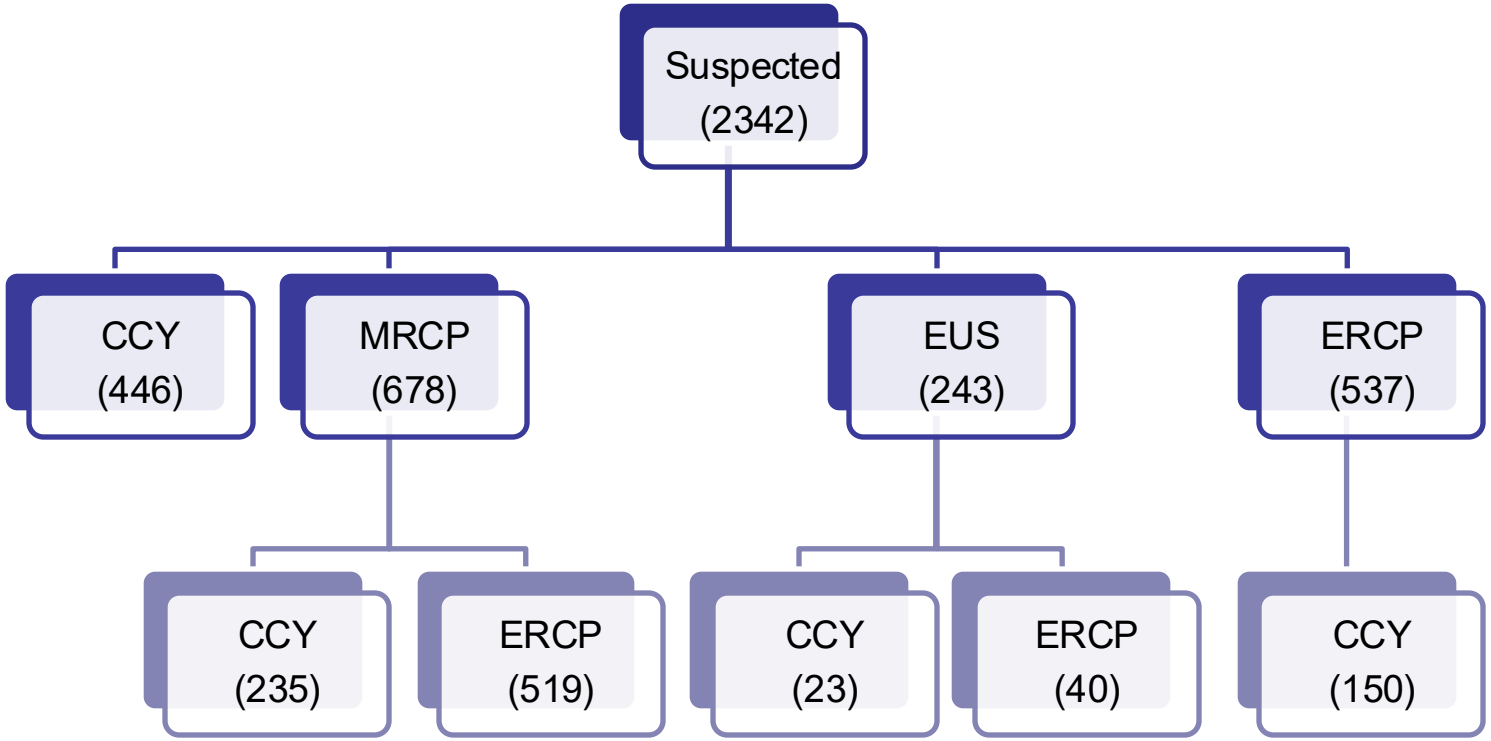
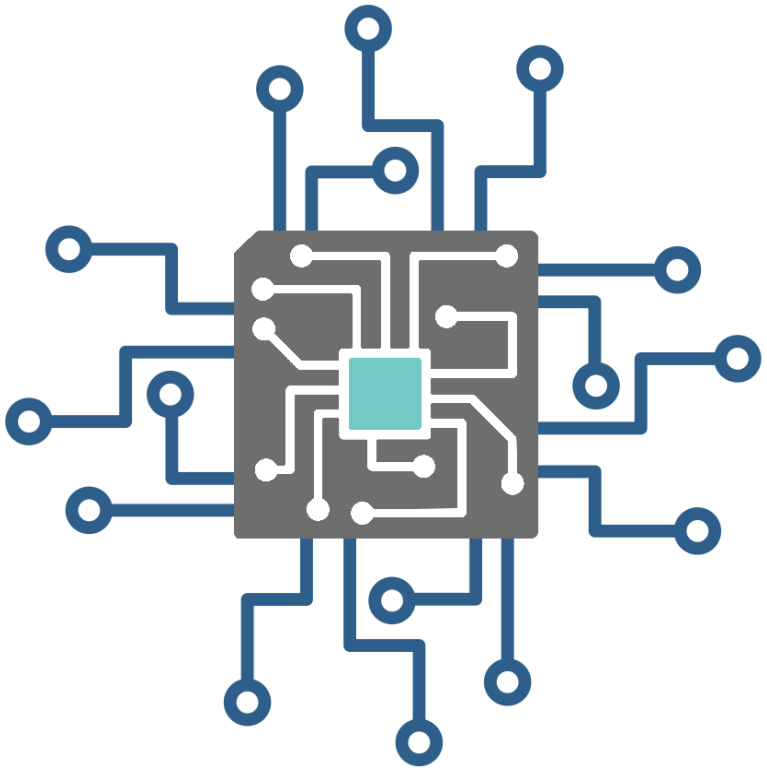


1103 confirmed choledocholithiasis

CCY = cholecystectomy, ERCP = endoscopic retrograde cholangiopancreatography,
EUS = endoscopic ultrasound, MRCP = magnetic resonance cholangiopancreatography

Prediction Model Creation



- 80/20 training/testing split
- Tested multiple iterations of model to balance performance and ease of use

Algorithm Choice and Variable Selection

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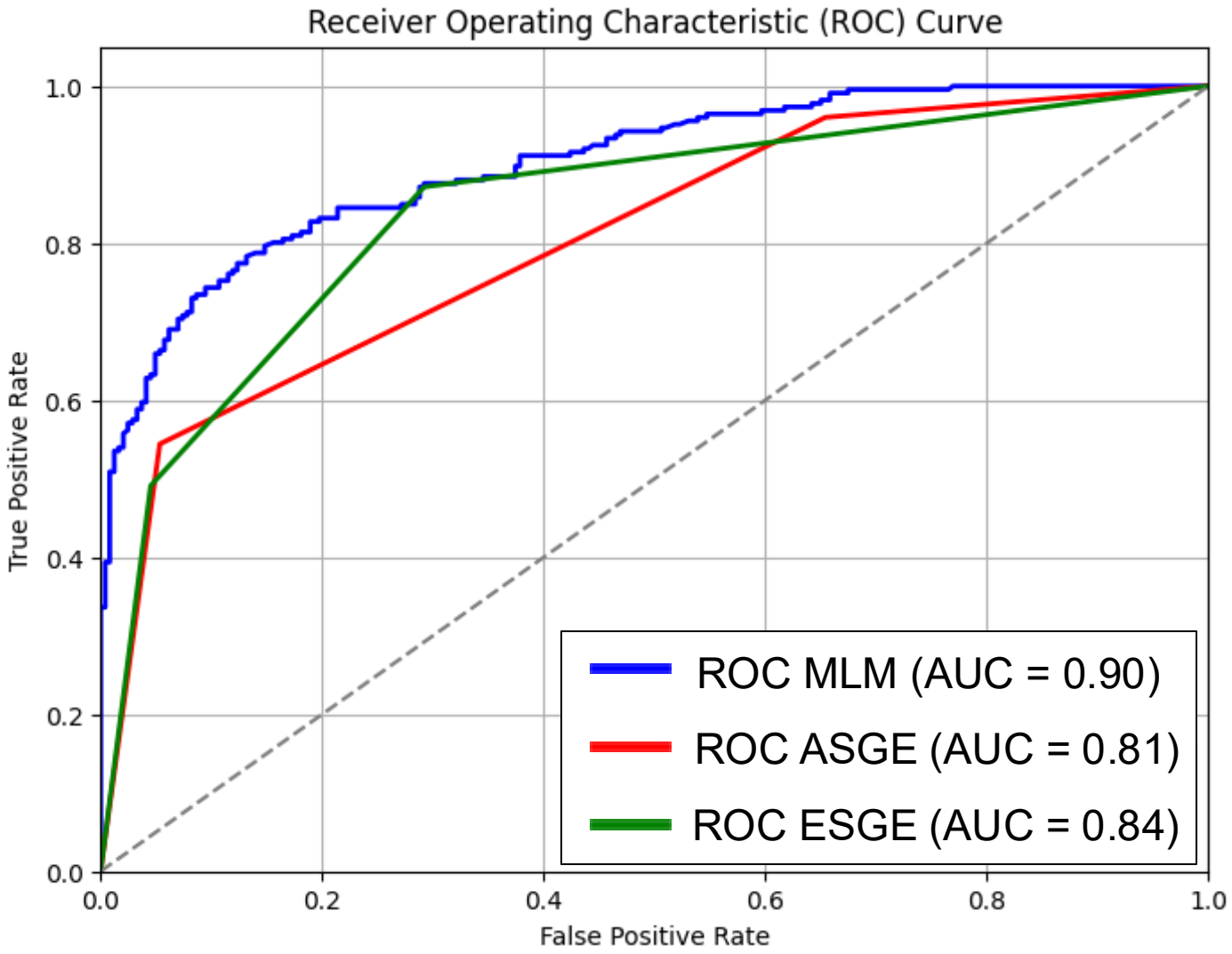
- Tested multiple iterations of the model to determine optimal variable selection and model performance
- Gradient boosting model (GBM*) algorithm performed better than logistic regression
- Trended lab values did not significantly enhance test performance
- Selection of mostly-observed variables prevents introduction of bias with missing value estimation
- Small amounts of missing data were estimated with an expectation maximization algorithm

Included Variables	Number of variables	Accuracy (%)		Precision (%)		Sensitivity (%)	
		W.o.I	EM	W.o.I	EM	W.o.I	EM
Mostly observed*	13	82	83	84	87	77	75
All relevant*	25	83	85	85	85	80	83
Excluding knowledge of pancreatitis, cholecystitis*	19	80	84	81	85	77	81
Excluding sex, 1 set of follow-up labs*	14	80	81	81	81	75	81
Excluding both sets of follow-up labs*	10	80	81	82	82	76	77
Logistic regression	10	78	78	82	81	70	70

W.o.I: Without imputation | EM: Expectation maximization for missing values

Model Performance

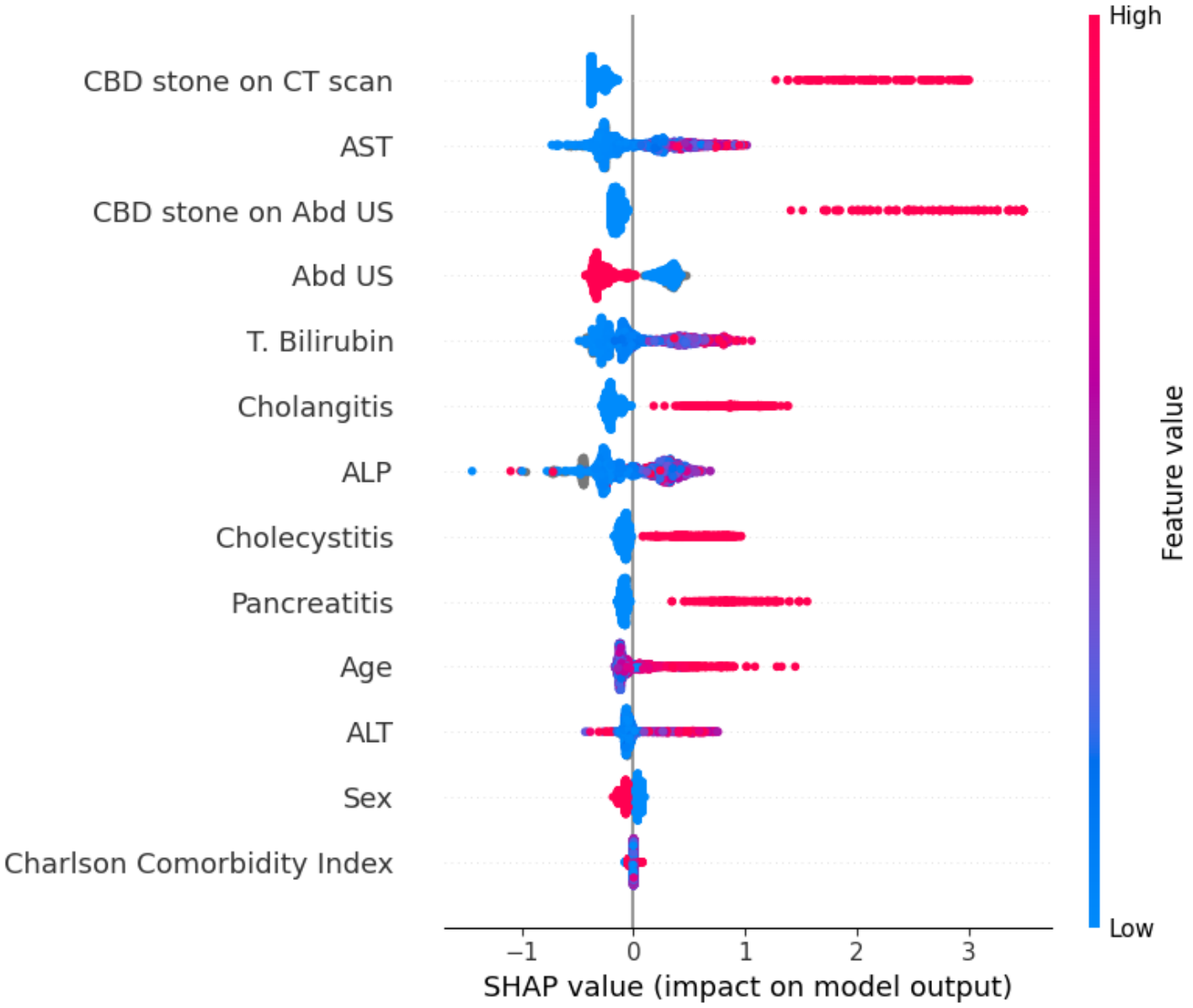


- The machine learning prediction model out-performed the existing guidelines' predictions, with an AUC of 0.90
- Increased precision prevents 85% of unnecessary ERCPs vs. ASGE guidelines and 66% vs. ESGE guidelines

	Accuracy	Precision	Recall
MLM Model	83%	87%	75%
ASGE 2019	65%	58%	96%
ESGE 2019	79%	74%	87%

Variable Importance: SHAP Analysis

- Of the mostly-observed variables, visualization of a CBD stone on imaging most strongly impacted model prediction



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1. Accurately predict the risk of choledocholithiasis
2. Guide appropriate intervention selection

Management Guidance

- Intuitive, web-based user interface allows real-time choledocholithiasis prediction using readily-available data
- Initial probability of choledocholithiasis is indexed against similar-risk patients to determine most frequently prescribed interventions

Patient Prediction Dashboard

Patient Description

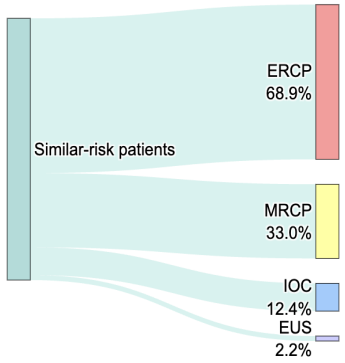
Sex	Female
Age	30
Cholangitis	NO
Pancreatitis	NO
Cholecystitis	YES
AST	210
ALT	645
ALP	110
T. Bilirubin	0.7
Abd US done?	☑
CBD stone on US	NO
CBD stone on CT	NO
Charlson Index	0

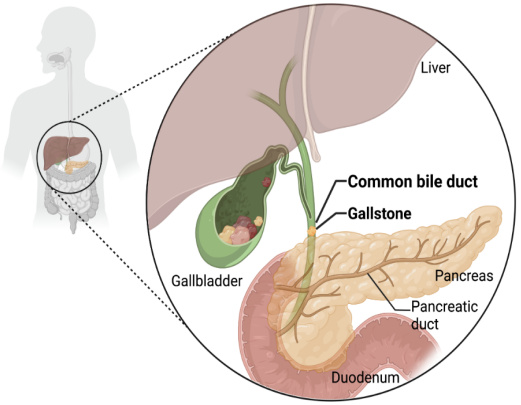
Clear | Calculate

Prediction

The probability of a stone in the CBD is 60.53%

Procedures Prescribed for Similar-Risk Patients





Cost Associated with Interventions

Test	Probability-Weighted Cost
IOC	$3294.19
MRCP	$3406.82
ERCP	$4451.00
EUS	$5145.64

Dash

Not Secure jupyter.dasl.jhsph.edu:1035

Patient Prediction Dashboard

Patient Description

Sex

Age

Cholangitis NO

Pancreatitis NO

Cholecystitis NO

AST

ALT

ALP

T. Bilirubin

Abd US done?

CBD stone on US NO

CBD stone on CT NO

Charlson Index

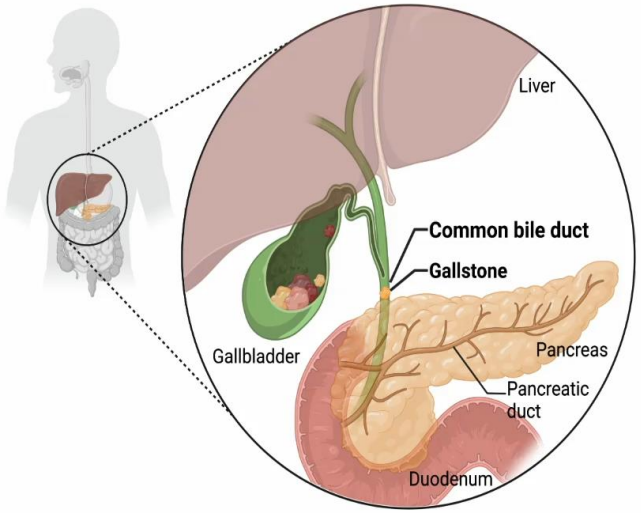
Clear Calculate

Prediction

Enter patient information and click 'Calculate' to view the results.

Procedures Prescribed for Similar-Risk Patients

Enter patient information and click 'Calculate' to view the results.



Cost Associated with Interventions

Test	Probability-Weighted Cost

Helpful Calculators:

Tokyo Guidelines for Acute Cholangitis (2018)

Charlson Comorbidity Index (CCI)

Patient Prediction Dashboard

Patient Description

Sex	Female
Age	30
Cholangitis	NO
Pancreatitis	NO
Cholecystitis	YES
AST	210
ALT	645
ALP	110
T. Bilirubin	3.9
Abd US done?	☑
CBD stone on US	NO
CBD stone on CT	NO
Charlson Index	0

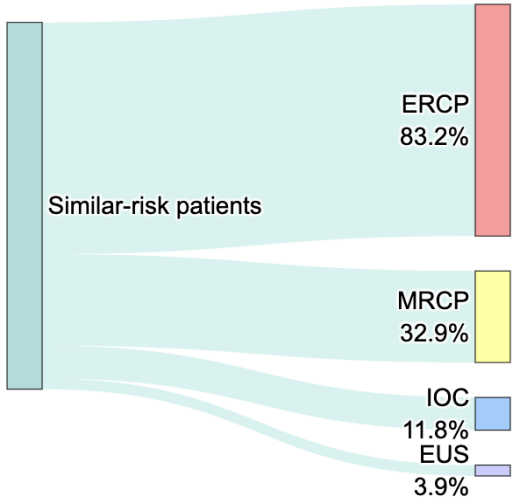
Clear

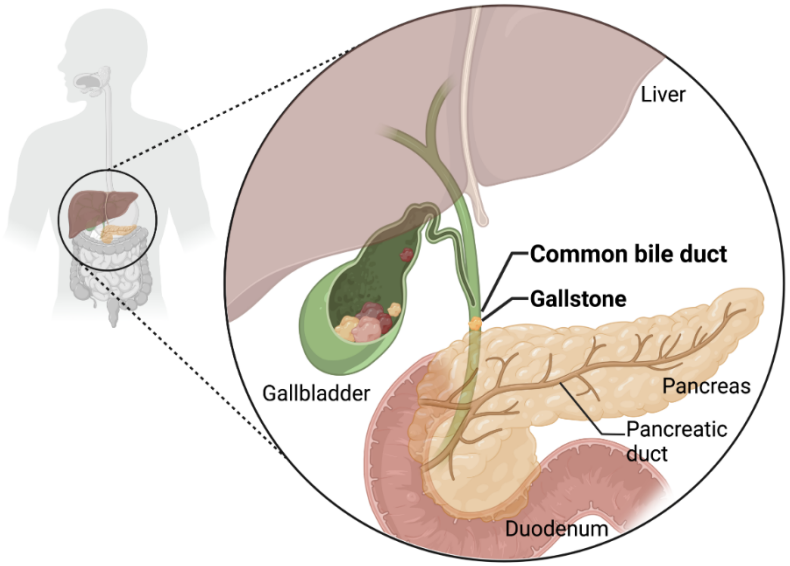
Calculate

Prediction

The probability of a stone in the CBD is 90.42%

Procedures Prescribed for Similar-Risk Patients





Cost Associated with Interventions

Test	Probability-Weighted Cost
IOC	$4624.59
MRCP	$4489.12
ERCP	$4451.00
EUS	$6211.79

Patient Prediction Dashboard

Patient Description

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Cholangitis	NO
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Cholecystitis	YES
AST	210
ALT	645
ALP	110
T. Bilirubin	0.7
Abd US done?	☑
CBD stone on US	NO
CBD stone on CT	NO
Charlson Index	0

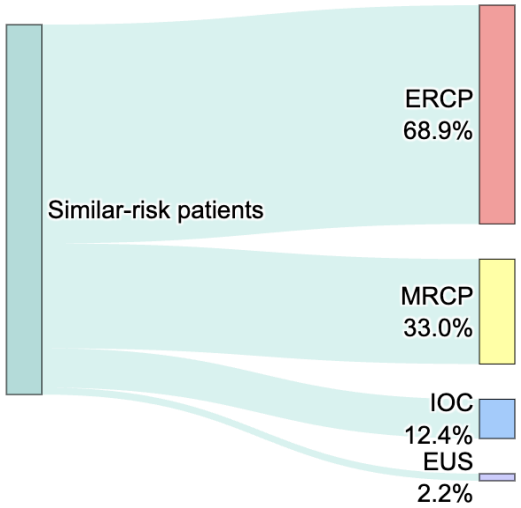
Clear

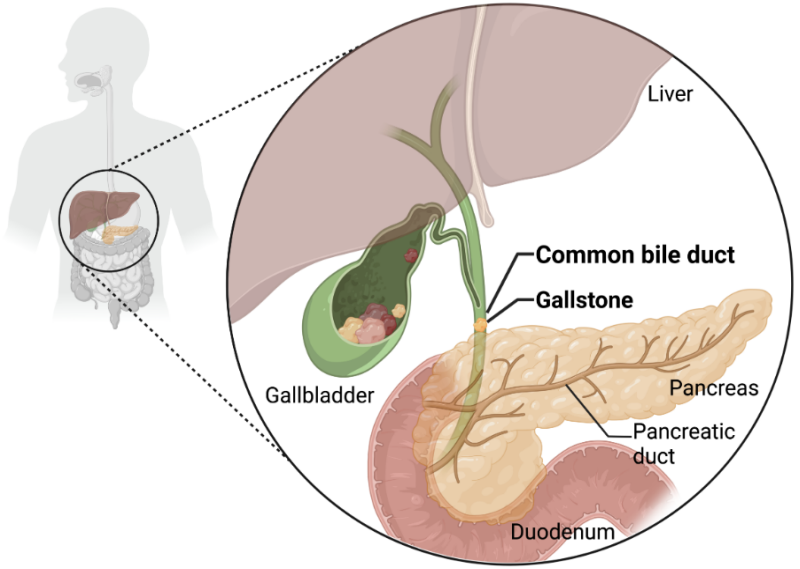
Calculate

Prediction

The probability of a stone in the CBD is 60.53%

Procedures Prescribed for Similar-Risk Patients





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Test	Probability-Weighted Cost
IOC	$3294.19
MRCP	$3406.82
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EUS	$5145.64

Limitations

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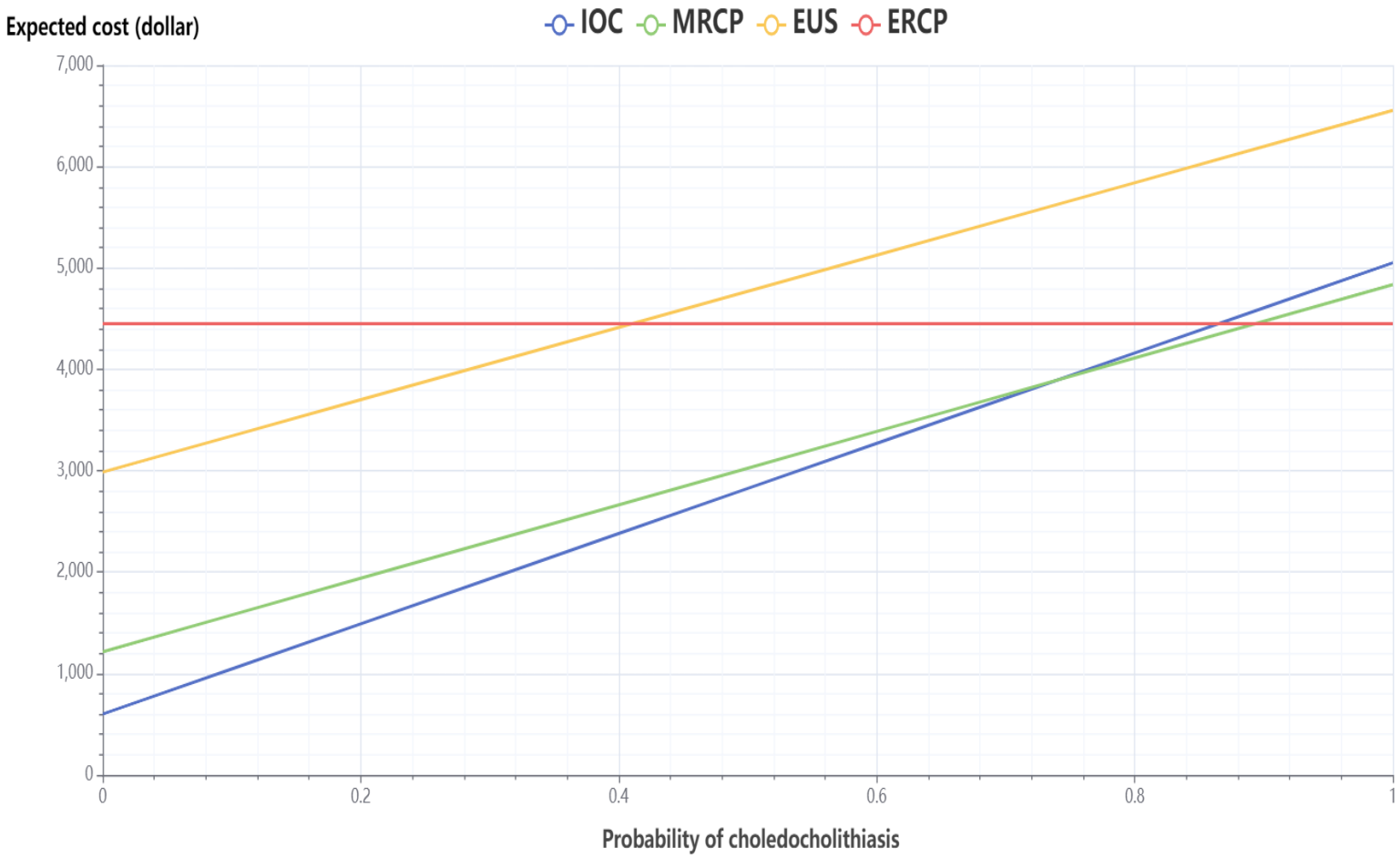
- Insufficient availability of follow-up lab values precluded analysis of dynamic liver enzyme trends
- Prioritization of enhanced precision contributes to increased false negatives
- Data availability is influenced by prescribing practices; limited set of missing data was imputed
- Suggested interventions are based on prescribing practices in our database; may not reflect the most efficient, efficacious, or cost-effective care strategy

Future Directions: Cost-Effectiveness Modeling and Validation



Probability of choledocholithiasis	Most cost-effective intervention*
0 – 74.1%	IOC
>74.1 – 89.37%	MRCP
>89.37 – 100%	ERCP

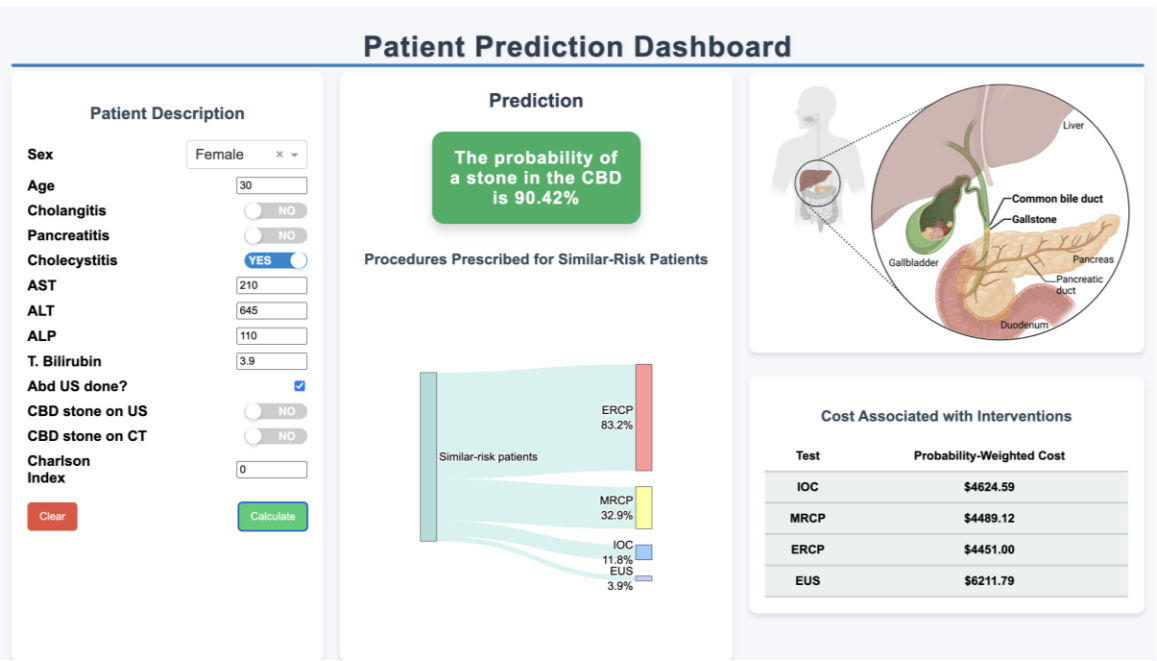
*Based on Medicare claims data: $307 MRCP, $2078.5 EUS, $4151 ERCP, $600 IOC

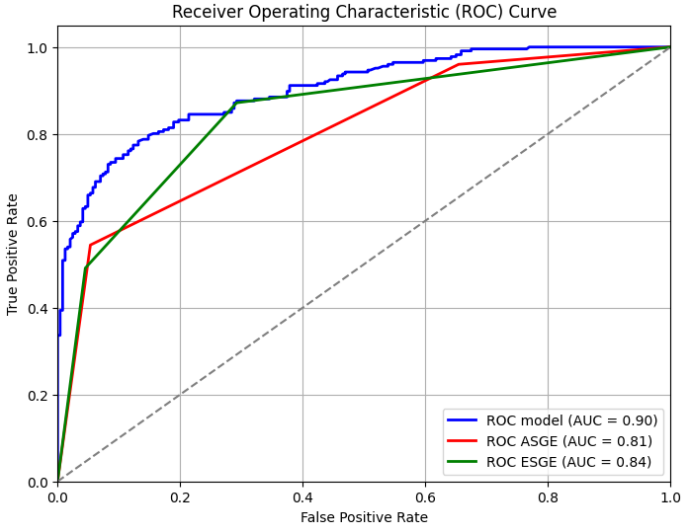
*Future models will incorporate non-procedural costs: anesthesia, hospital stay, complications, etc.

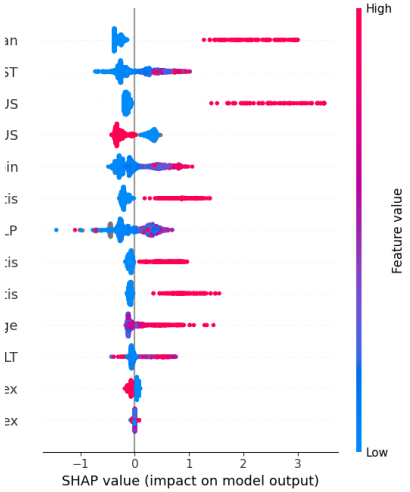
CCY = cholecystectomy, ERCP = endoscopic retrograde cholangiopancreatography, EUS = endoscopic ultrasound, IOC = intraoperative cholangiogram, MRCP = magnetic resonance cholangiopancreatography

Conclusions

- Machine-learning decision tool offers individualized risk assessment for choledocholithiasis using readily available clinical information
- Our model enhances predictive ability compared to existing guidelines
- Risk stratification guides management, limiting unnecessary ERCPs
- This tool may reduce procedural morbidity and promote cost-effective, high-value care for patients with biliary disease







Acknowledgements

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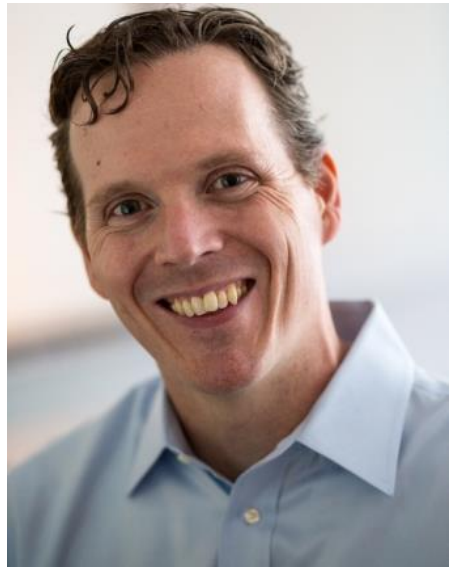

QUESTIONS

APPENDIX

Algorithm Choice and Variable Selection

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- Tested multiple iterations of the model to determine optimal variable selection and model performance

Included Variables	Number of variables	Test accuracy (%)			Precision (%)			Sensitivity (%)		
		W.I	W.o.I	EM	W.I	W.o.I	EM	W.I	W.o.I	EM
Mostly observed	13	84	82	83	88	84	87	77	77	75
All	25	84	83	85	83	85	85	82	80	83
Excluding knowledge of pancreatitis, cholecystitis	19	83	80	84	83	81	85	80	77	81
Excluding sex, 1 set of follow-up labs	14	83	80	81	85	81	81	80	75	81
Excluding both sets of follow-up labs	10	83	80	81	85	82	82	78	76	77
Logistic regression	10	80	78	78	84	82	81	71	70	70

W.I: With an indicator for missing values | W.o.I: Without an indicator for missing values | EM: Expectation maximization for missing values

Algorithm Choice and Variable Selection

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Mostly observed	13	84	82	83	88	84	87	77	77	75
All	25	84	83	85	83	85	85	82	80	83
Excluding knowledge of pancreatitis, cholecystitis	19	83	80	84	83	81	85	80	77	81
Excluding sex, 1 set of follow-up labs	14	83	80	81	85	81	81	80	75	81
Excluding both sets of follow-up labs	10	83	80	81	85	82	82	78	76	77
Logistic regression	10	80	78	78	84	82	81	71	70	70

W.I: With an indicator for missing values | W.o.I: Without an indicator for missing values | EM: Expectation maximization for missing values

Algorithm Choice and Variable Selection

JOHNS HOPKINS MEDICINE

- Tested multiple iterations of the model to determine optimal variable selection and model performance
- Gradient boosting algorithm performed better than logistic regression
- Trended lab values did not significantly enhance test performance
- Selection of mostly-observed variables prevents introduction of bias with missing value estimation

Included Variables	Number of variables	Test accuracy (%)			Precision (%)			Sensitivity (%)		
		W.I	W.o.I	EM	W.I	W.o.I	EM	W.I	W.o.I	EM
Mostly observed	13	84	82	83	88	84	87	77	77	75
All	25	84	83	85	83	85	85	82	80	83
Excluding knowledge of pancreatitis, cholecystitis	19	83	80	84	83	81	85	80	77	81
Excluding sex, 1 set of follow-up labs	14	83	80	81	85	81	81	80	75	81
Excluding both sets of follow-up labs	10	83	80	81	85	82	82	78	76	77
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Algorithm Choice and Variable Selection

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- Tested multiple iterations of the model to determine optimal variable selection and model performance
- Gradient boosting algorithm performed better than logistic regression
- Trended lab values did not significantly enhance test performance
- Selection of mostly-observed variables prevents introduction of bias with missing value estimation
- Small amounts of missing data were estimated with an expectation maximization algorithm

Included Variables	Number of variables	Test accuracy (%)			Precision (%)			Sensitivity (%)		
		W.I	W.o.I	EM	W.I	W.o.I	EM	W.I	W.o.I	EM
Mostly observed	13	84	82	83	88	84	87	77	77	75
All	25	84	83	85	83	85	85	82	80	83
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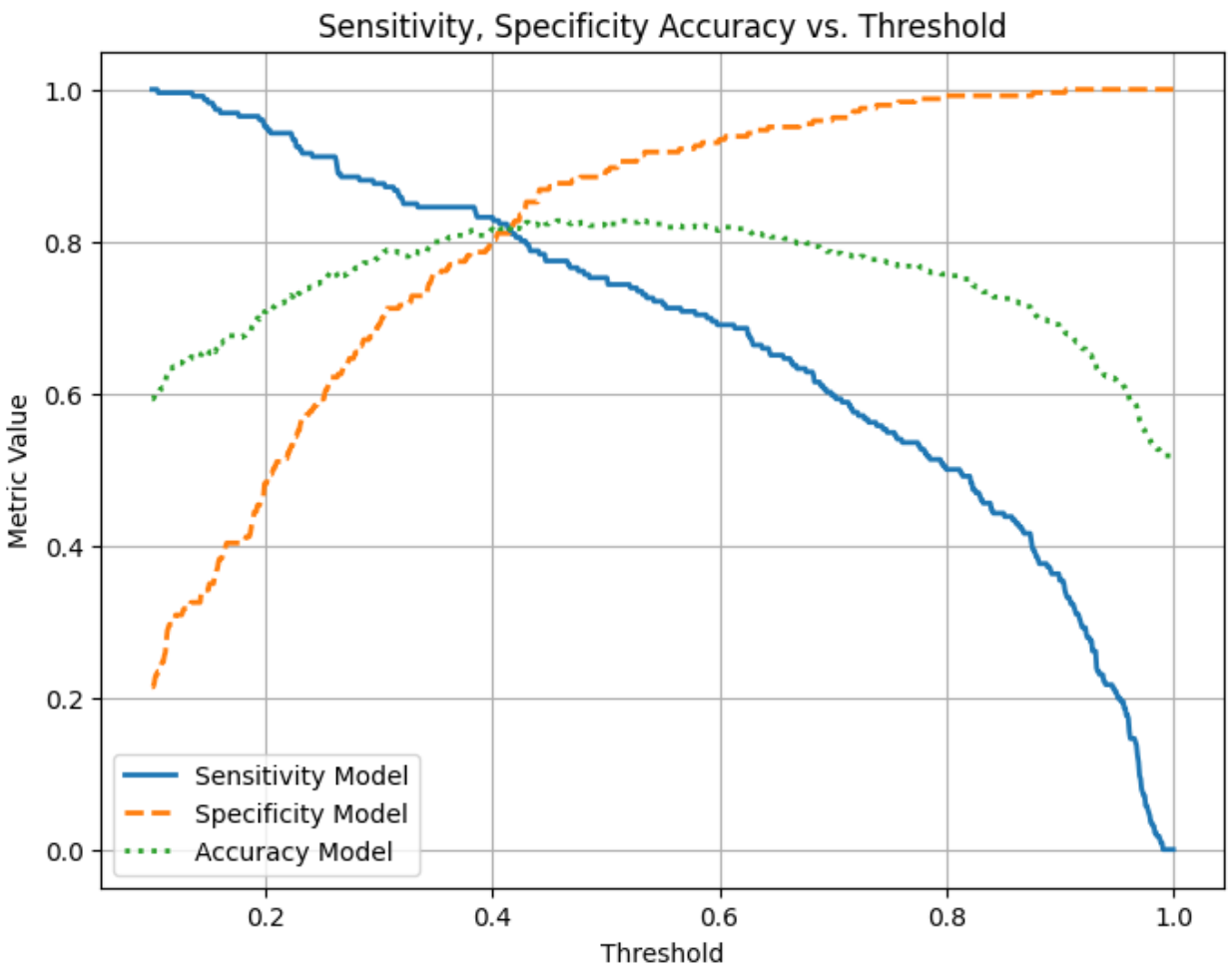
Future Directions: Trended Labs

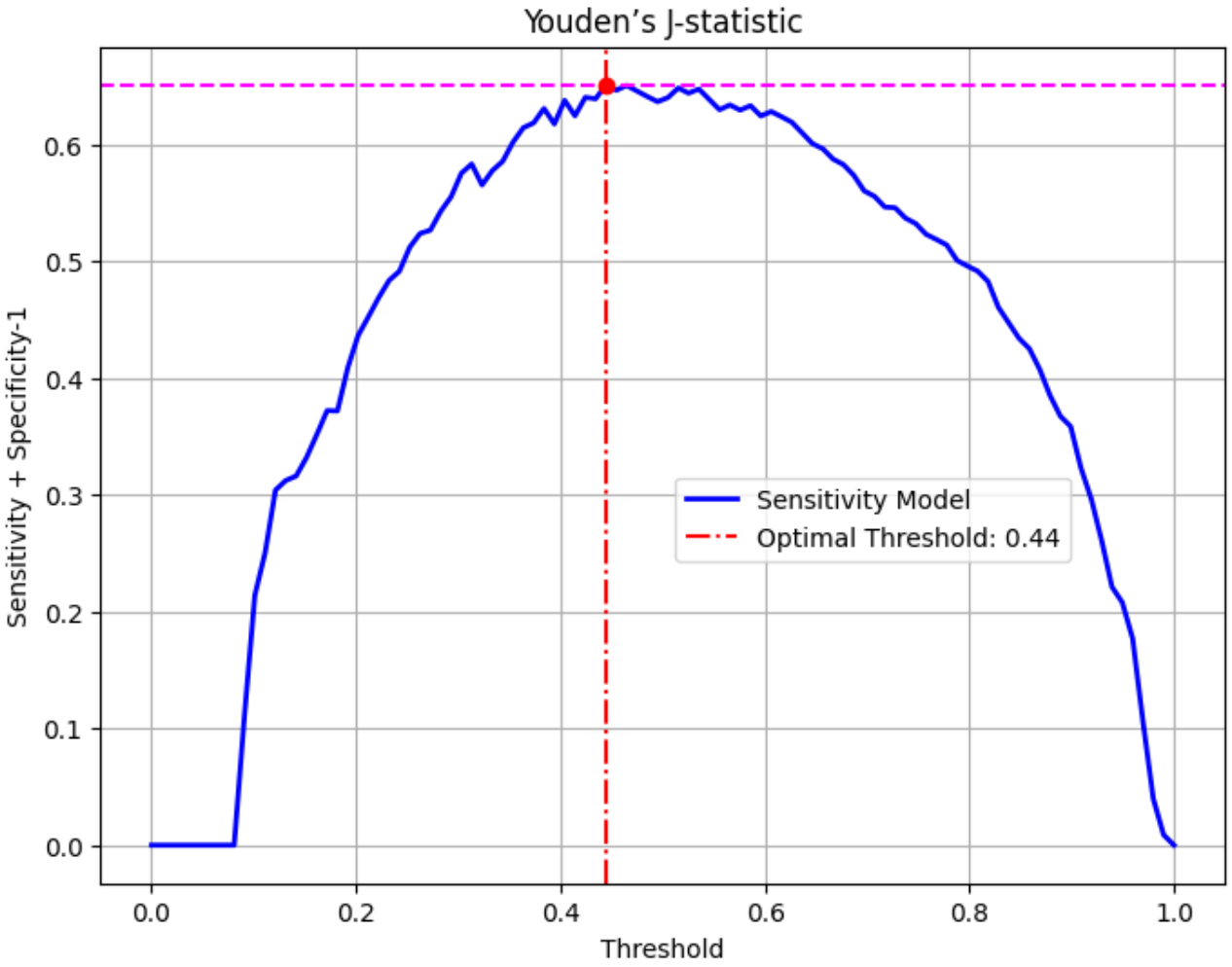
JOHNS HOPKINS MEDICINE

	# features	Accuracy	Precision	Recall
Logistic regression without follow-up labs	13	76	74	95
Logistic regression with follow-up labs	17	73	71	92

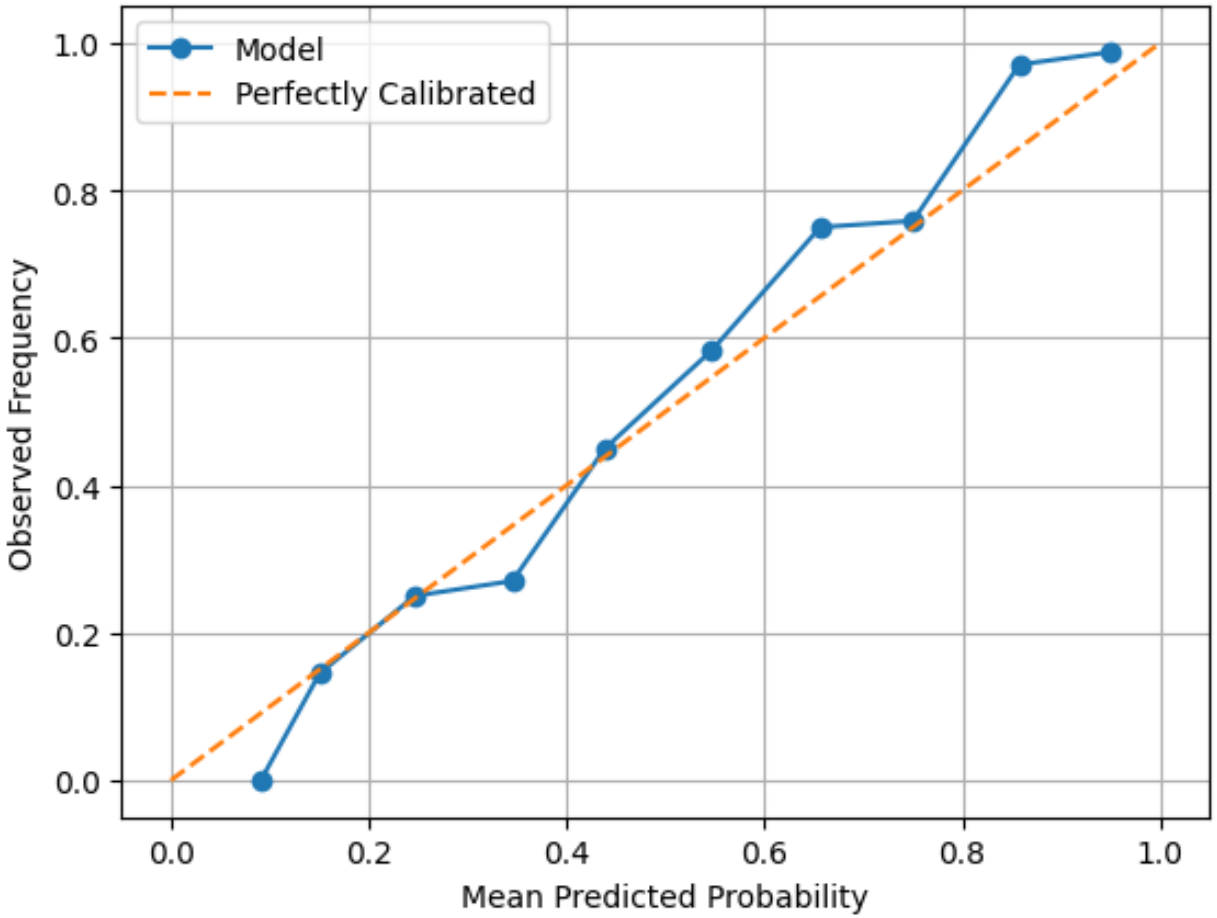
CCY = cholecystectomy, ERCP = endoscopic retrograde cholangiopancreatography, EUS = endoscopic ultrasound, IOC = intraoperative cholangiogram, MRCP = magnetic resonance cholangiopancreatography

Threshold selection





Reliability Diagram (Calibration Curve)

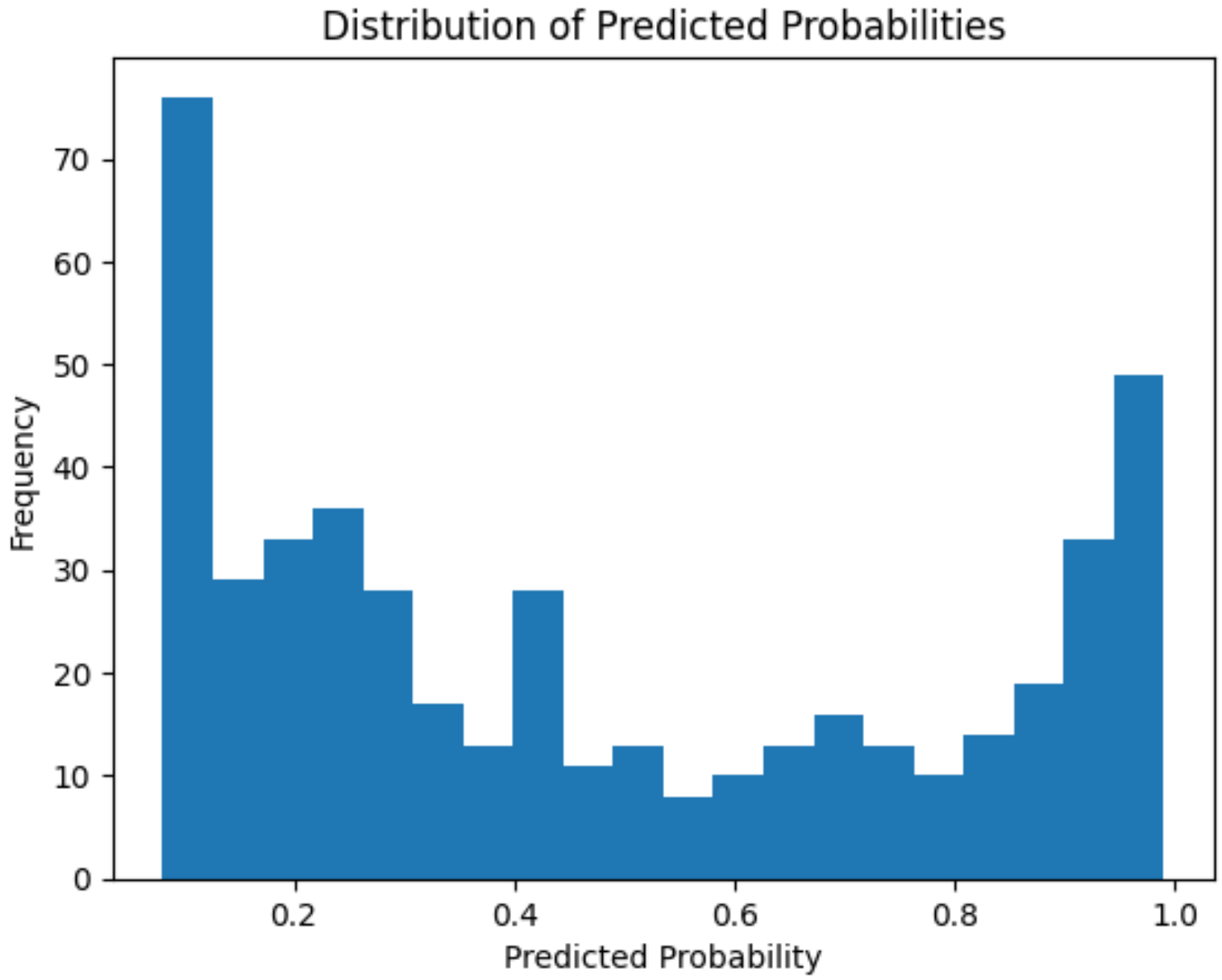


The Brier score is 0.004, representing the mean squared error between the predicted probabilities and the actual outcomes. It measures overall accuracy and calibration.

Performance Comparison

JOHNS HOPKINS MEDICINE

Model	False Positive Rate	False Negative Rate
The Model	9.9%	25.0%
ASGE 2019	65.4%	4.0%
ESGE 2019	29.2%	12.8%


Distribution of Predicted Probabilities
Frequency
0
10
20
30
40
50
60
70
0.2
0.4
0.6
0.8
1.0
Predicted Probability